

PROJECT BOATY

# Subsystem Specification Helm

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

|          |                                                                            |
|----------|----------------------------------------------------------------------------|
| Document | BOATY-SSS-HLM                                                              |
| Issue    | Issue E (for review)                                                       |
| Date     | 30 September 2026                                                          |
| Status   | For review by the project owner                                            |
| Parents  | SRS Issue G, ADD Issue G, ICD Issue G, FMEA Issue F, KCL Issue C           |
| Content  | 35 allocated SRS requirements → 44 subsystem requirements (39 M, 3 S, 2 C) |

# Document control

| Issue | Date              | Change                                                                                                                                                                                                                     | By                                            |
|-------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| A     | 28 September 2026 | First issue, for review.                                                                                                                                                                                                   | Claude (drafted)                              |
| B     | 28 September 2026 | CR-02: microSD logging (HLM-D02). FMEA actions A-01, A-02, A-06, A-18: HLM-D24 formalised; HLM-D39 to D41 added; parameter baseline extended.                                                                              | Claude, owner decisions (CR-02, FMEA actions) |
| C     | 28 September 2026 | CR-03: board class and real price (Matek F405-TE class, £62) in budget and HLM-D02.                                                                                                                                        | Claude, owner decision                        |
| D     | 29 September 2026 | CR-04: SpeedyBee F405 WING APP (HLM-D02). CR-05: GCS failsafe 3 s (HLM-D27). Simulator results folded into HLM-D05/08/20/24/25/26/28/29; HLM-D42 to D44 added; parameter baseline now generated from software/params.      | Claude, owner decisions                       |
| E     | 30 September 2026 | SDR WP2: HLM-D19 met by Dijkstra path planning with slide avoidance (OA_TYPE, OA_MARGIN_MAX, AVOID_BEHAVE; FENCE_MARGIN 3 → 2 m), verified by SC-27. HLM-D09: HDOP gate moved to Mission Control (not available in Rover). | Claude                                        |

| Role                | Name                      | Signature / date |
|---------------------|---------------------------|------------------|
| Author              | Claude, for Project Boaty |                  |
| Reviewer / approver | Project owner             |                  |

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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# 1. Scope

## 1.1 Purpose

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

## 1.2 Boundary

| Inside this subsystem                                                                                                                                                                                                                                                                                                | Outside (and who owns it)                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Flight controller hardware (SpeedyBee F405 WING APP + PDB)</li><li>• ArduPilot Rover firmware (third party) and the controlled parameter set</li><li>• GNSS + compass module and mast cabling</li><li>• Fence and mission storage</li><li>• Dataflash / SD logging</li></ul> | <ul style="list-style-type: none"><li>• Everything that sends commands (MCN via IF-02, MCP via IF-04)</li><li>• Motor drives (PRP)</li><li>• Power sources (PWR)</li></ul> |

## 1.3 Breakdown

| Item                    | Description                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| HLM-1 Flight controller | SpeedyBee F405 WING APP (STM32F405, ChibiOS), PDB with V/I sense, microSD                                |
| HLM-2 Firmware          | ArduPilot Rover 4.7.1, pinned                                                                            |
| HLM-3 Parameter set     | software/params/boaty-mk1.parm (behaviour, also flown in SITL) + boaty-mk1-speedybee.parm (board wiring) |
| HLM-4 GNSS/compass      | u-blox M10 + QMC5883L on the mast                                                                        |

## 2. Applicable documents

| Document               | Use                                                           |
|------------------------|---------------------------------------------------------------|
| BOATY-SRS-001 Issue D  | Parent requirements                                           |
| BOATY-ADD-001 Issue D  | Allocation, decisions, budgets, early verification items      |
| BOATY-ICD-001 Issue C  | Interface definitions (all listed in section 3 apply in full) |
| BOATY-FMEA-001 Issue B | Failure modes and actions carried into this issue             |

## 3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

| Interface                         | Other end             | Role     | ICD |
|-----------------------------------|-----------------------|----------|-----|
| <b>IF-02</b> Command & telemetry  | Mission Control       | Conforms | \$4 |
| <b>IF-04</b> Companion link       | Mission computer      | Conforms | \$4 |
| <b>IF-05</b> Motor commands       | Propulsion            | Owner    | \$6 |
| <b>IF-07</b> Helm power & sensing | Power                 | Conforms | \$6 |
| <b>IF-09</b> Beacon drive         | Recovery & signalling | Owner    | \$6 |
| <b>IF-19</b> Mast socket          | Hull & structure      | Conforms | \$8 |
| <b>IF-22</b> GNSS & compass       | internal              | Owner    | \$6 |

## 4. Parent requirements

**Primary (35):** this subsystem is responsible for meeting and verifying these SRS requirements.

| SRS            | Pri      | Requirement (abridged)                                                                                                   | Met by           |
|----------------|----------|--------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>MOD-001</b> | <b>M</b> | The system shall implement exactly these boat modes: DISARMED, HOLD, MANUAL, AUTO and RTL, as defined in section 2.3.    | HLM-D05          |
| <b>MOD-002</b> | <b>M</b> | On power-up the boat shall enter DISARMED, with both motors inhibited.                                                   | HLM-D04          |
| <b>MOD-004</b> | <b>M</b> | The system shall refuse to arm unless every pre-arm check (PRE-001 to PRE-008) passes, and shall state the failing...    | HLM-D08          |
| <b>MOD-005</b> | <b>M</b> | When an AUTO mission completes, the boat shall return home inside the fence, avoiding exclusions (RTL behaviour,...      | HLM-D06          |
| <b>MOD-008</b> | <b>C</b> | The boat shall disarm automatically after 60 s in HOLD at home with no operator input, and announce it.                  | HLM-D07          |
| <b>PRE-001</b> | <b>M</b> | GNSS shall have a 3D fix with at least 8 satellites and HDOP no greater than 1.5.                                        | HLM-D09          |
| <b>PRE-002</b> | <b>M</b> | A geofence shall be loaded and enabled, and home shall lie inside the inclusion fence and outside all exclusion zones.   | HLM-D10          |
| <b>PRE-003</b> | <b>M</b> | Battery state of charge shall be at least 80%.                                                                           | HLM-D11          |
| <b>PRE-004</b> | <b>M</b> | The heading sensor shall report healthy and calibrated.                                                                  | HLM-D12          |
| <b>PRE-007</b> | <b>M</b> | The independent safety layer (SAF-001) shall report healthy.                                                             | HLM-D13          |
| <b>PRE-008</b> | <b>M</b> | Logging shall be active on the helm and Mission Control, each with at least 1 GB free.                                   | HLM-D02, HLM-D14 |
| <b>NAV-002</b> | <b>M</b> | Cruise speed in calm water shall be 1.0 m/s $\pm$ 0.2 m/s.                                                               | HLM-D15          |
| <b>NAV-003</b> | <b>M</b> | Speed through the water shall be limited to 1.5 m/s in every mode.                                                       | HLM-D16          |
| <b>NAV-004</b> | <b>M</b> | In AUTO the boat shall pass within the acceptance radius of each waypoint. The radius is configurable from 1 to 10 m,... | HLM-D17          |
| <b>NAV-005</b> | <b>S</b> | Cross-track error between waypoints shall be no more than 3 m RMS in wind up to 3 m/s.                                   | HLM-D18, HLM-D42 |
| <b>NAV-006</b> | <b>S</b> | In HOLD with station-keeping, the boat shall stay within 5 m of the hold point in wind up to 3 m/s.                      | HLM-D18          |
| <b>NAV-007</b> | <b>M</b> | In RTL the boat shall navigate to home without entering any exclusion zone or leaving the inclusion fence.               | HLM-D19          |
| <b>NAV-008</b> | <b>S</b> | Above 0.3 m/s the heading estimate shall use GNSS course as well as the magnetometer, so a disturbed compass cannot...   | HLM-D20          |
| <b>FEN-001</b> | <b>M</b> | The system shall support one inclusion polygon of 3 to 70 vertices and up to 10 exclusion zones, each a polygon or...    | HLM-D10, HLM-D21 |
| <b>FEN-004</b> | <b>M</b> | The fence shall be enforced in every armed mode, including MANUAL.                                                       | HLM-D22          |
| <b>FEN-005</b> | <b>M</b> | On a fence breach the boat shall enter RTL within 1 s of the breach.                                                     | HLM-D23          |
| <b>FEN-007</b> | <b>M</b> | Fence and exclusion zones shall not be changed while the boat is armed.                                                  | HLM-D25          |
| <b>FS-001</b>  | <b>M</b> | Low battery: when estimated state of charge falls to 35%, the boat shall enter RTL. At 15% it shall continue RTL at...   | HLM-D26, HLM-D40 |
| <b>FS-002</b>  | <b>M</b> | Link loss in MANUAL: with no command link for 3 s the boat shall enter HOLD, and after 10 s it shall enter RTL. (3 s...  | HLM-D27          |
| <b>FS-003</b>  | <b>M</b> | Link loss in AUTO: the boat shall continue the mission. If the link is still lost after 60 s it shall enter RTL.         | HLM-D27          |
| <b>FS-005</b>  | <b>M</b> | Stuck detection: if commanded thrust is at least 50% and ground speed stays below 0.1 m/s for 5 s, the boat shall...     | HLM-D29          |
| <b>FS-007</b>  | <b>M</b> | Mission computer failure: if the helm receives no mission computer heartbeat for 3 s, it shall complete the loaded...    | HLM-D30          |

| SRS            | Pri      | Requirement (abridged)                                                                                                   | Met by                    |
|----------------|----------|--------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>FS-011</b>  | <b>M</b> | When several failsafes are active, the most conservative response shall win: motors stopped over RTL over continuing...  | HLM-D31                   |
| <b>FS-012</b>  | <b>M</b> | Every failsafe event shall be logged, and announced at Mission Control within 2 s if the link is up.                     | HLM-D32                   |
| <b>SAF-001</b> | <b>M</b> | Fence and failsafe functions shall run on a processor separate from the mission computer and Mission Control. That...    | HLM-D01, HLM-D02, HLM-D03 |
| <b>SAF-002</b> | <b>M</b> | Those functions shall keep working with the mission computer, Mission Control and the radio link all failed.             | HLM-D30, HLM-D33          |
| <b>SAF-003</b> | <b>M</b> | The mission computer and Mission Control shall command the helm only through the helm interface (SWE-002). They shall... | HLM-D34, HLM-D44          |
| <b>COM-007</b> | <b>C</b> | An independent manual-override radio link (e.g. ExpressLRS) may be added.                                                | HLM-D37                   |
| <b>MEC-013</b> | <b>M</b> | The GNSS and compass shall be mounted at least 150 mm from motor and power wiring.                                       | HLM-D35, HLM-D41          |
| <b>LOG-001</b> | <b>M</b> | The helm shall log position, attitude, mode, commands, battery and failsafe events at 5 Hz or faster.                    | HLM-D36                   |

**Supporting (24):** another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-003, MOD-006, PRE-005, NAV-001, FEN-006, FS-004, FS-008, FS-009, FS-010, SAF-007, MIS-001, MIS-002, MIS-003, MIS-004, MIS-005, MIS-006, DET-004, MC-014, REC-004, PWR-005, PWR-006, PWR-010, SWE-008, ENV-002

## 5. Subsystem requirements

Requirements are numbered HLM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

### 5.1 Hardware and firmware

| ID      | Requirement                                                                                                                                                                                                                                                                                                                                                                                                        | Pri | Ver | Stage | Trace                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|--------------------------------------|
| HLM-D01 | The helm shall run a pinned ArduPilot Rover stable release built for the chosen board: Rover 4.7.1 (tag Rover-4.7.1, commit db79216), the release the simulator flies. Version and hash are recorded in the repository.                                                                                                                                                                                            | M   | I   | BENCH | SAF-001, SWE-006                     |
| HLM-D02 | The flight controller shall provide: $\geq 2$ free UARTs (GNSS, companion), $\geq 3$ DShot-capable outputs, $\geq 2$ analogue battery inputs, I2C, IMU and baro, and a microSD slot ( $\geq 8$ GB card) for logging (CR-02), and run from $\geq 7$ V so it survives pack sag near empty. Baseline: SpeedyBee F405 WING APP (7-36 V, 5 usable UARTs, 12 DShot outputs, microSD; CR-04), wireless module not fitted. | M   | I   | BENCH | SAF-001, PRE-008, V-01, DD-15, DD-19 |
| HLM-D03 | No project-written code (including Lua) shall run on the flight controller. It runs ChibiOS only.                                                                                                                                                                                                                                                                                                                  | M   | I   | BENCH | SAF-001, DD-13                       |
| HLM-D04 | On power-up the helm shall be disarmed with motor outputs at neutral. Auto-arming shall be disabled.                                                                                                                                                                                                                                                                                                               | M   | T   | BENCH | MOD-002                              |

### 5.2 Modes

| ID      | Requirement                                                                                                                                                                                                                                                        | Pri | Ver | Stage | Trace          |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|----------------|
| HLM-D05 | Only STEERING, HOLD, LOITER, AUTO, RTL and GUIDED (weed-shedding only) shall be used. The helm shall boot and arm into HOLD (INITIAL_MODE 4), with no RC mode switch (MODE_CH 0) and any RC receiver ignored (RC_OPTIONS 1). Without these SITL armed into MANUAL. | M   | T   | SIM   | MOD-001, DD-12 |
| HLM-D06 | At mission end the helm shall RTL and then hold at home. The mission-done behaviour is set to hold as a backstop.                                                                                                                                                  | M   | S   | SIM   | MOD-005, V-04  |
| HLM-D07 | Auto-disarm after 60 s at home is performed by Mission Control (MCN-D12). The helm shall accept it.                                                                                                                                                                | C   | S   | SIM   | MOD-008        |

### 5.3 Pre-arm checks

| ID      | Requirement                                                                                                                                                                                                         | Pri | Ver | Stage | Trace                  |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|------------------------|
| HLM-D08 | All ArduPilot arming checks shall be enabled except the RC-channels check (no receiver is fitted: ARMING_SKIPCHK 64), and failures reported as STATUSTEXT (MCN translates them).                                    | M   | S   | SIM   | MOD-004                |
| HLM-D09 | Arming shall need a 3D fix and a healthy EKF. Rover has no HDOP arming gate (GPS_HDOP_GOOD is Copter-only; SDR RID-11), so the HDOP $\leq 1.5$ and $\geq 8$ satellite checks are done by Mission Control (MCN-D15). | M   | T   | BENCH | PRE-001                |
| HLM-D10 | Arming shall need the fence enabled and loaded, with home inside it.                                                                                                                                                | M   | S   | SIM   | PRE-002, FEN-001       |
| HLM-D11 | Arming shall need $\geq 80\%$ battery capacity remaining.                                                                                                                                                           | M   | S   | SIM   | PRE-003                |
| HLM-D12 | Arming shall need calibrated, consistent compass and EKF.                                                                                                                                                           | M   | T   | BENCH | PRE-004                |
| HLM-D13 | Arming shall be refused with motor-rail voltage $< 9.0$ V (key out).                                                                                                                                                | M   | T   | BENCH | PRE-007, MOD-003, V-13 |
| HLM-D14 | Arming shall need logging available.                                                                                                                                                                                | M   | T   | BENCH | PRE-008                |

### 5.4 Navigation

| ID             | Requirement                                                                                                                                                                                                                                                                                                            | Pri      | Ver  | Stage | Trace            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|-------|------------------|
| <b>HLM-D15</b> | Cruise speed shall be set to 1.0 m/s and tuned to $\pm 0.2$ m/s.                                                                                                                                                                                                                                                       | <b>M</b> | T    | POOL  | NAV-002          |
| <b>HLM-D16</b> | Waypoint and RTL speeds shall be $\leq 1.5$ m/s. Maximum throttle shall be limited so full throttle in STEERING gives $\leq 1.5$ m/s in calm water.                                                                                                                                                                    | <b>M</b> | T    | POOL  | NAV-003          |
| <b>HLM-D17</b> | Waypoint acceptance radius shall default to 3 m (configurable 1-10 m).                                                                                                                                                                                                                                                 | <b>M</b> | S    | SIM   | NAV-004          |
| <b>HLM-D18</b> | Steering and speed controllers shall be tuned to $\leq 3$ m RMS cross-track error and $\leq 5$ m loiter error in 3 m/s wind.                                                                                                                                                                                           | <b>S</b> | T    | LAKE  | NAV-005, NAV-006 |
| <b>HLM-D19</b> | RTL shall plan around exclusion zones inside the inclusion fence: ArduPilot's Dijkstra planner (OA_TYPE 2, OA_MARGIN_MAX 6 m) with fence avoidance set to slide along a zone's margin (AVOID_BEHAVE 0, FENCE_MARGIN 2 m). With Rover's default 'stop' behaviour the planned path stalled the boat (SC-27, SDR RID-01). | <b>M</b> | S, T | SIM   | NAV-007, V-01    |
| <b>HLM-D20</b> | NAV-008 shall be met by the helm's EKF together with B7's first-motion heading check (MCP-D22). V-16 showed the EKF alone keeps a wrong heading with its single compass rotated 90°, though navigation still tracked on GNSS course.                                                                                   | <b>S</b> | S    | SIM   | NAV-008, V-16    |

## 5.5 Fence

| ID             | Requirement                                                                                                                                                                                                                                                   | Pri      | Ver | Stage | Trace                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-------|-----------------------------------|
| <b>HLM-D21</b> | The helm shall hold one inclusion polygon ( $\leq 70$ vertices) and $\leq 10$ exclusion polygons or circles, plus a 100 m circle backstop about home at Milton.                                                                                               | <b>M</b> | S   | SIM   | FEN-001, OPS-006                  |
| <b>HLM-D22</b> | The fence shall be enforced in every armed mode used.                                                                                                                                                                                                         | <b>M</b> | S   | SIM   | FEN-004, V-02                     |
| <b>HLM-D23</b> | A breach shall trigger RTL within 1 s.                                                                                                                                                                                                                        | <b>M</b> | S   | SIM   | FEN-005                           |
| <b>HLM-D24</b> | Persistent breach ( $> 30$ s or $> 10$ m outside) shall stop the motors. V-14 found no native mechanism, so B7 (MCP-D30) and Mission Control (MCN-D59) each independently command HOLD, and either suffices. The helm's fence RTL stays the primary response. | <b>M</b> | S   | SIM   | FEN-006, V-14, A-18, FM-14, DD-21 |
| <b>HLM-D25</b> | Fence changes while armed shall be refused by every sender: C7 (IF-14) and the IF-04 filter. V-17 showed the helm itself accepts them, so these two are the only enforcement.                                                                                 | <b>M</b> | S   | SIM   | FEN-007, V-17                     |

## 5.6 Failsafes

| ID             | Requirement                                                                                                                                                                                                                     | Pri      | Ver | Stage | Trace                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-------|-----------------------------------|
| <b>HLM-D26</b> | Battery failsafe: RTL at 35% remaining; at 15%, continue RTL and alarm. V-15 found no native speed reduction, so B6 lowers the RTL speed (MCP-D29).                                                                             | <b>M</b> | S   | SIM   | FS-001, V-15                      |
| <b>HLM-D27</b> | GCS failsafe on system-255 heartbeats: FS_GCS_TIMEOUT 2 s + FS_TIMEOUT 1 s = HOLD 3 s after the link goes (CR-05; both are the firmware minimums), continue in AUTO. The later RTL steps are done by MCP B4 (MCP-D12).          | <b>M</b> | T   | SIM   | FS-002, FS-003, V-03, V-06, DD-22 |
| <b>HLM-D28</b> | EKF/position failsafe $\rightarrow$ HOLD as a backstop. It took 9 s in SITL, so B6 stops the motors within FS-004's 3 s (MCP-D28) and does the recovery RTL (MCP-D15).                                                          | <b>M</b> | S   | SIM   | FS-004, DD-21                     |
| <b>HLM-D29</b> | Crash/stuck check $\rightarrow$ HOLD (CRASH_THR_MIN 50%, CRASH_VEL_MIN 0.1 m/s, CRASH_TIMEOUT 5 s), raising an event the MCP acts on. One noisy GNSS speed sample resets its timer (V-05), so B7's second detector backs it up. | <b>M</b> | S   | POOL  | FS-005, V-05                      |
| <b>HLM-D30</b> | No companion-computer failsafe shall be configured: the helm completes missions without the MCP.                                                                                                                                | <b>M</b> | S   | SIM   | FS-007, SAF-002                   |
| <b>HLM-D31</b> | When failsafes combine, motors-stopped outranks RTL, which outranks continue. Demonstrated in SITL combinations.                                                                                                                | <b>M</b> | S   | SIM   | FS-011                            |

| ID      | Requirement                                                | Pri | Ver | Stage | Trace  |
|---------|------------------------------------------------------------|-----|-----|-------|--------|
| HLM-D32 | Every failsafe shall be logged and reported by STATUSTEXT. | M   | S   | SIM   | FS-012 |

## 5.7 Independence, identity and logging

| ID      | Requirement                                                                                                                                         | Pri | Ver | Stage | Trace                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|-----------------------|
| HLM-D33 | All fence and failsafe functions shall work with the MCP, Mission Control and the link all absent.                                                  | M   | S   | SIM   | SAF-002               |
| HLM-D34 | Only system 255 is the GCS. Parameter writes are accepted only while disarmed through IF-14. Boat services are filtered (IF-04).                    | M   | T   | SIM   | SAF-003, IF-02, IF-04 |
| HLM-D35 | The GNSS/compass shall be mast-mounted $\geq 150$ mm from power wiring, with motor interference $\leq 30\%$ after a motor-interference calibration. | M   | T   | BENCH | MEC-013               |
| HLM-D36 | The helm shall log position, attitude, mode, commands, battery and failsafes at $\geq 5$ Hz.                                                        | M   | T   | BENCH | LOG-001               |
| HLM-D37 | A spare UART shall be kept for an optional ExpressLRS receiver, with RC failsafe = RTL if fitted.                                                   | C   | I   | BENCH | COM-007               |
| HLM-D38 | The beacon output shall be configured per IF-09.                                                                                                    | S   | T   | BENCH | IF-09, REC-004        |

## 5.8 FMEA-driven (Issue B)

| ID      | Requirement                                                                                                                                                                                                       | Pri | Ver | Stage | Trace                              |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|------------------------------------|
| HLM-D39 | GNSS glitch rejection (EKF3, glitch radius $\leq 25$ m) and GNSS data-timeout detection shall be enabled. A glitch or frozen position shall not produce an uncommanded exit from the fence (SC-20, SC-21, L2-04). | M   | S   | SIM   | FS-004, FS-013, A-01, FM-02, FM-03 |
| HLM-D40 | Voltage thresholds shall back up the mAh battery failsafe: low 10.2 V and critical 9.6 V under load for $\geq 10$ s, with the same actions. Whichever triggers first applies.                                     | M   | T   | BENCH | FS-001, A-06, FM-15                |
| HLM-D41 | Compass interference shall be measured on rig L2 at full thrust before the hull layout is frozen. Above 30%, wiring or mast height changes before build.                                                          | M   | T   | BENCH | MEC-013, A-02, FM-04               |

## 5.9 Found in simulation (Issue D)

| ID      | Requirement                                                                                                                                                                                                                                 | Pri | Ver | Stage | Trace                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|-----------------------|
| HLM-D42 | The throttle-to-thrust curve shall be linearised (MOT_THST_EXPO), because props give thrust $\sim \text{throttle}^2$ . At the default a 180° pivot took 20 s in SITL; at 1.0 it takes 6 s. Re-tuned from thrust-stand data before the pool. | M   | T   | POOL  | NAV-001, NAV-005      |
| HLM-D43 | Motor power shall be limited (BATT_WATT_MAX 70 W) so pack sag near empty cannot brown out the ESCs or flight controller (KCL section 5), while leaving thrust to make headway in the ENV-002 wind.                                          | M   | A   | BENCH | PWR-005, DD-19        |
| HLM-D44 | The helm shall have exactly one MAVLink command path: SERIAL1 to the mission computer. The flight controller's wireless module is not fitted and SERIAL6 is disabled.                                                                       | M   | I   | BENCH | SAF-003, IF-02, IF-04 |

## 6. Controlled parameter baseline

The configuration that implements the requirements above. **This table is generated from software/params/boaty-mk1.parm (behaviour, also flown in the simulator) and boaty-mk1-speedybee.parm (board wiring)**, so it is always the file that is flashed. Every behaviour parameter was checked against Rover 4.7.1 in SITL (IF-14-02).

| Group                               | Parameter       | Value | Why                                                                                                                                              |
|-------------------------------------|-----------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Vehicle                             | FRAME_CLASS     | 2     | Boat (MOD-001, V-01)                                                                                                                             |
| Motors: skid steer, outputs 1 and 4 | SERVO1_FUNCTION | 73    | ThrottleLeft                                                                                                                                     |
| Motors: skid steer, outputs 1 and 4 | SERVO2_FUNCTION | 0     |                                                                                                                                                  |
| Motors: skid steer, outputs 1 and 4 | SERVO3_FUNCTION | 0     |                                                                                                                                                  |
| Motors: skid steer, outputs 1 and 4 | SERVO4_FUNCTION | 74    | ThrottleRight                                                                                                                                    |
| Motors: skid steer, outputs 1 and 4 | MOT_THR_MAX     | 100   | model top speed ~1.5 m/s (NAV-003); re-tune in pool                                                                                              |
| Motors: skid steer, outputs 1 and 4 | MOT_THST_ASYM   | 2.67  | 2.0 N ahead / 0.75 N astern per pod (PRP-D02)                                                                                                    |
| Motors: skid steer, outputs 1 and 4 | MOT_THST_EXPO   | 1.0   | props: thrust ~ throttle <sup>2</sup> . At 0 a 180 deg pivot took 20 s in SITL, at 1.0 it takes 6 s. Re-tune from thrust-stand data (KCL DS-09). |
| Motors: skid steer, outputs 1 and 4 | MOT_REV_DELAY   | 0.2   | s, ESC reversal pause (KCL KC-03; tune on L2)                                                                                                    |
| Motors: skid steer, outputs 1 and 4 | MOT_BAT_WATT_TC | 2     | s, power-limit time constant                                                                                                                     |
| Speeds and navigation               | CRUISE_SPEED    | 1.0   | m/s (NAV-002)                                                                                                                                    |
| Speeds and navigation               | CRUISE_THROTTLE | 70    | % for 1.0 m/s in the model; measure in the pool                                                                                                  |
| Speeds and navigation               | WP_SPEED        | 1.0   | m/s (NAV-004)                                                                                                                                    |
| Speeds and navigation               | WP_RADIUS       | 3     | m                                                                                                                                                |
| Speeds and navigation               | RTL_SPEED       | 0     | 0 = WP_SPEED (NAV-007)                                                                                                                           |
| Speeds and navigation               | LOIT_RADIUS     | 2     | m (NAV-006)                                                                                                                                      |
| Speeds and navigation               | MIS_DONE_BEHAVE | 0     | Hold at mission end (MOD-005 backstop)                                                                                                           |
| Speeds and navigation               | INITIAL_MODE    | 4     | arm into HOLD, never MANUAL (found in SITL)                                                                                                      |
| Speeds and navigation               | MODE_CH         | 0     | no RC mode switch: it overrode INITIAL_MODE                                                                                                      |
| Speeds and navigation               | RC_OPTIONS      | 1     | ignore any RC receiver: none is fitted (IF-02)                                                                                                   |
| Identity                            | MAV_SYSID       | 1     |                                                                                                                                                  |
| Identity                            | MAV_GCS_SYSID   | 255   | only Mission Control counts as the GCS (V-06)                                                                                                    |
| Fence                               | FENCE_ENABLE    | 1     |                                                                                                                                                  |
| Fence                               | FENCE_TYPE      | 6     | circle (2) + polygon (4)                                                                                                                         |
| Fence                               | FENCE_ACTION    | 1     | RTL or Hold                                                                                                                                      |

| Group                             | Parameter        | Value | Why                                                                                                                                                                                 |
|-----------------------------------|------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fence                             | FENCE_RADIUS     | 100   | m backstop circle (OPS-006)                                                                                                                                                         |
| Fence                             | FENCE_MARGIN     | 2     | m: fence avoidance keeps this clear. Below the validator's 3 m leg clearance, and well below OA_MARGIN_MAX (SC-27: at 3 m it stalled RTL)                                           |
| Path planning<br>round exclusions | OA_TYPE          | 2     | Dijkstra                                                                                                                                                                            |
| Path planning<br>round exclusions | AVOID_BEHAVE     | 0     | slide along a zone's margin, don't stop. The planner's legs can pass close to a zone; with Rover's default (stop) RTL deadlocked (SC-27)                                            |
| Path planning<br>round exclusions | OA_MARGIN_MAX    | 6     | m round zones for planned paths. Must exceed FENCE_MARGIN plus the ~2 m the boat cuts each corner, or fence avoidance stalls the boat on the planned path (SC-27; ArduPilot #11601) |
| Battery                           | BATT_MONITOR     | 4     | analogue voltage + current                                                                                                                                                          |
| Battery                           | BATT_CAPACITY    | 2500  | mAh; PWR-D20 minimum, replace with measured                                                                                                                                         |
| Battery                           | BATT_LOW_MAH     | 875   | 35% remaining -> RTL                                                                                                                                                                |
| Battery                           | BATT_CRT_MAH     | 375   | 15% remaining -> RTL                                                                                                                                                                |
| Battery                           | BATT_FS_LOW_ACT  | 1     | RTL                                                                                                                                                                                 |
| Battery                           | BATT_FS_CRT_ACT  | 1     | RTL                                                                                                                                                                                 |
| Battery                           | BATT_ARM_MAH     | 2000  | 80% needed to arm (PRE-003)                                                                                                                                                         |
| Battery                           | BATT_LOW_VOLT    | 10.2  | V, voltage backstop (HLM-D40)                                                                                                                                                       |
| Battery                           | BATT_CRT_VOLT    | 9.6   | V                                                                                                                                                                                   |
| Battery                           | BATT_FS_VOLTSRC  | 1     | sag-compensated voltage (KCL KF-01)                                                                                                                                                 |
| Battery                           | BATT_WATT_MAX    | 70    | W, keeps the rail above 7.5 V near empty (KF-02)                                                                                                                                    |
| Motor-rail sense:<br>arming key   | BATT2_MONITOR    | 3     | analogue voltage only                                                                                                                                                               |
| Motor-rail sense:<br>arming key   | BATT2_VOLT_PIN   | 15    | AIRSPD pad on the SpeedyBee; voltage2 in SITL                                                                                                                                       |
| Motor-rail sense:<br>arming key   | BATT2_ARM_VOLT   | 9.0   | V: no arming with the key out                                                                                                                                                       |
| Failsafes                         | FS_GCS_ENABLE    | 2     | enabled, continue mission in AUTO (FS-002/003)                                                                                                                                      |
| Failsafes                         | FS_GCS_TIMEOUT   | 2     | s without a system-255 heartbeat (minimum 2)                                                                                                                                        |
| Failsafes                         | FS_TIMEOUT       | 1     | s the failsafe must persist (minimum 1). Total ~3 s: FS-002 relaxed to 3 s (CR-05).                                                                                                 |
| Failsafes                         | FS_ACTION        | 2     | Hold                                                                                                                                                                                |
| Failsafes                         | FS_THR_ENABLE    | 0     | no RC receiver fitted (manual is via IF-02)                                                                                                                                         |
| Failsafes                         | FS_EKF_ACTION    | 1     | Hold (FS-004)                                                                                                                                                                       |
| Failsafes                         | FS_CRASH_CHECK   | 1     | Hold (FS-005)                                                                                                                                                                       |
| Failsafes                         | CRASH_THR_MIN    | 50    | % throttle ...                                                                                                                                                                      |
| Failsafes                         | CRASH_VEL_MIN    | 0.1   | ... with speed below this (FS-005) ...                                                                                                                                              |
| Failsafes                         | CRASH_TIMEOUT    | 5     | ... for this long -> Hold (FS-005: <= 5 s)                                                                                                                                          |
| Failsafes                         | EK3_GLITCH_RAD   | 25    | m (FS-004, HLM-D39)                                                                                                                                                                 |
| Arming                            | ARMING_SKIPCHK   | 64    | skip only the RC-channels check: no RC receiver                                                                                                                                     |
| Board wiring                      | SERIAL1_PROTOCOL | 2     | MAVLink 2 to the Pi Zero (IF-04)                                                                                                                                                    |
| Board wiring                      | SERIAL1_BAUD     | 115   |                                                                                                                                                                                     |
| Board wiring                      | SERIAL2_PROTOCOL | -1    | unused (SBUS inverter on RX)                                                                                                                                                        |
| Board wiring                      | SERIAL3_PROTOCOL | 5     | GPS (IF-22)                                                                                                                                                                         |

| Group        | Parameter        | Value | Why                                   |
|--------------|------------------|-------|---------------------------------------|
| Board wiring | SERIAL6_PROTOCOL | -1    | wireless board not fitted (KCL KF-04) |
| Board wiring | MOT_PWM_TYPE     | 6     | DShot300 (IF-05)                      |
| Board wiring | SERVO_DSHOT_ESC  | 1     | AM32                                  |
| Board wiring | SERVO_BLH_3DMASK | 9     | outputs 1 and 4 reversible            |
| Board wiring | BATT_VOLT_PIN    | 10    |                                       |
| Board wiring | BATT_CURR_PIN    | 11    |                                       |
| Board wiring | BATT_VOLT_MULT   | 11.05 | calibrate on the bench (KF-10)        |
| Board wiring | BATT_AMP_PERVLT  | 50    | calibrate on the bench (KF-10)        |
| Board wiring | BATT2_VOLT_MULT  | 11.0  | 10k / 1k divider (IF-06)              |
| Board wiring | LOG_BACKEND_TYPE | 1     | microSD                               |
| Board wiring | NTF_LED_TYPES    | 257   | board LEDs + NeoPixel                 |
| Board wiring | SERVO12_FUNCTION | 120   | NeoPixel1 on the LED pad (IF-09)      |
| Board wiring | GPS_TYPE         | 1     | auto                                  |
| Board wiring | COMPASS_EXTERNAL | 1     |                                       |

## 6.1 Where each failsafe step lives

| Requirement | Helm (native)                      | Boat Python (MCP)                                    | Mission Control              |
|-------------|------------------------------------|------------------------------------------------------|------------------------------|
| FS-001      | RTL at 35%; RTL + alarm at 15%     | Slow RTL to 0.6 m/s at 15% (B6)                      | Announce                     |
| FS-002      | HOLD at 3 s (CR-05)                | RTL at 10 s (B4)                                     | Show 'link lost'             |
| FS-003      | Continue in AUTO                   | RTL at 60 s (B4)                                     | -                            |
| FS-004      | EKF failsafe HOLD (backstop, ~9 s) | HOLD within 3 s; RTL after 10 s healthy (B6)         | Announce                     |
| FS-005/006  | HOLD on stuck (crash check)        | Second detector at 10 s (B7); ≤ 3 astern bursts (B5) | Announce                     |
| FEN-005/006 | RTL on breach                      | HOLD at 30 s or 10 m outside (B7)                    | HOLD if persistent (MCN-D59) |

Rule: nothing outside the helm is the last line of defence for an action the helm can take. Where the helm cannot (FS-001 speed, FS-004 timing, FEN-006) the boat services add the step, and each is proven in simulation (software/results).

## 7. Design constraints and budgets

- ADD DD-19 (CR-04): SpeedyBee F405 WING APP, 7-36 V input
- ADD DD-13: no Lua; Python on the MCP covers Lua-type behaviours
- No project-written code on the FC (SAF-001)

| Budget          | Allocation                                              |
|-----------------|---------------------------------------------------------|
| Mass allocation | ≤ 45 g (estimate 39 g: FC + PDB 25 g, GNSS 14 g)        |
| Cost allocation | £63 (SpeedyBee F405 WING APP £45, GNSS £14, microSD £4) |
| Power           | ≈ 0.8 W including GNSS                                  |

## 8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

| Stage | Test | Demo | Inspect | Analysis | Sim | Reqs      |
|-------|------|------|---------|----------|-----|-----------|
| SIM   | 4    | -    | -       | -        | 20  | <b>23</b> |
| BENCH | 10   | -    | 5       | 1        | -   | <b>16</b> |

| Stage | Test | Demo | Inspect | Analysis | Sim | Reqs     |
|-------|------|------|---------|----------|-----|----------|
| POOL  | 3    | -    | -       | -        | 1   | <b>4</b> |
| LAKE  | 1    | -    | -       | -        | -   | <b>1</b> |

| Ref                             | Open item                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------|
| V-01                            | Bench check of the SpeedyBee with Rover 4.7.1 (every feature was exercised in SITL)      |
| V-02 ... V-06, V-11<br>... V-17 | Answered in SITL (ADD section 9); results folded into the requirements below             |
| Tuning                          | MOT_THST_EXPO, CRUISE_THROTTLE, steering gains from thrust-stand and pool data (HLM-D42) |
| HLM-OI-1 / CR-02                | Closed: owner chose a flight controller with microSD (ADD DD-15)                         |